Approximate Bayesian Inference

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Approximate Bayesian inference

The General problem

- Observations D.
- Nuisance variables z.
- Unknown parameter θ.
- Direct calculation of any of these terms can be infeasible:

$$\beta(\theta \mid D) = \frac{P_\theta(D)\beta(\theta)}{\sum_{\theta'} P_{\theta'}(D)\beta(\theta')}, \qquad P_\theta(D) = \sum_z P_\theta(D, z).$$

Common methods

- Monte Carlo
- Variational Bayes
- Approximate Bayesian Computation (ABC)
- Stochastic Variational Inference

The Monte-Carlo sampler

To approximate an expectation, we can approximate the integral

$$\mathbb{E}_P(f) \triangleq \int_\Omega f(\omega) dP(\omega)$$

by sampling ω from P and taking the sample mean:

$$\mathbb{E}_P(f) \approx \frac{1}{N} \sum_{n=1}^{N} f(\omega^{(n)}),$$

which as we already know, is an unbiased estimator.
We can also approximate the probabilities of events $A \subset \Omega$:

$$P(A) = \int_A dP(\omega) = \int_\Omega \mathbb{I}\{\omega \in A\} \, dP(\omega) \tag{1}$$

$$= \mathbb{E}_P[\mathbb{I}\{\omega \in A\}] \approx \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}\left\{\omega^{(n)} \in A\right\}. \tag{2}$$

But what if we can not sample from P?

Importance sampling

If we cannot sample from P, but can sample from some other distribution Q, we can use this trick (statisticians hate it!). Let's write it first out for finite Ω, for simplicity

$$\mathbb{E}_P(f) \triangleq \sum_{\omega \in \Omega} f(\omega) \frac{P(\omega)}{Q(\omega)} Q(\omega) = \mathbb{E}_Q(fP/Q)$$

Now this is the expectation of $f(\omega)P(\omega)/Q(\omega)$ with respect to Q. So, if we can sample from Q instead, we can still approximate the integral for the general case.

$$\mathbb{E}_P(f) \triangleq \int_\Omega f(\omega) \frac{dP(\omega)}{dQ(\omega)} dQ(\omega) \approx \frac{1}{N} \sum_{n=1}^{N} f(\omega^{(n)}) \frac{dP(\omega^{(n)})}{dQ(\omega^{(n)})}, \qquad \omega^{(n)} \sim Q.$$

It is easy to show that this estimator is also unbiased. However, as the probability ratio can be very large, it has a very high variance.
So, how can we sample from P for arbitrary distributions?

Basic sampling theory

Inversion sampler

$F(u) \triangleq \mathbb{P}(x \geq u) = P(\{\omega : x(\omega) \geq u\})$ is the CDF of x.

- Sample u uniformly in $[0, 1]$
- Set $x = F^{-1}(u)$, i.e. $v : \mathbb{P}(x \geq v) = u$

Rejection Sampler

- Input: Threshold ϵ, distribution Q
- Repeat:
- $\hat{x} \sim Q$.
- $u \sim \mathrm{Unif}[0, 1]$
- Until $u \leq P(\hat{x})/\epsilon Q(\hat{x})$.
- Return $\hat{x}$

Notes

- Useful for sampling from a known distribution P.
- Indirectly useful from sampling from unknown distributions.

Monte-Carlo sampling

$$\beta(B \mid D) = \frac{\int_B P_\theta(D) d\beta(\theta)}{\int_\Theta P_{\theta'}(D)\beta(\theta')}$$

We can approximate the integrals by sampling from the prior β:

$$\int_B P_\theta(D) d\beta(\theta) \approx \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}\left\{\theta^{(n)} \in B\right\} P_{\theta^{(n)}}(D), \qquad \theta^{(n)} \sim \beta.$$

- Sampling from the prior is inefficient.
- The estimator has high bias and variance.
- So, we can use Markov Chain Monte Carlo. This lets us sample a sequence $\theta^{(n)}$ which converges asymptotically to $\beta(\theta^{(n)}|D)$.

Markov Chain Monte Carlo

MCMC for posterior sampling

- Form a Markov chain $P(\theta^{(n+1)} \mid \theta^{(n)}, D)$

MCMC for other latent variables

- Form a Markov chain $P(z^{(n+1)} \mid z^{(n)}, D)$

Metropolis-Hastings

Algorithm (symmetric version)

- Input: Proposal distribution $Q(x|x') = Q(x'|x)$
- At time n:
- $\hat{x} \sim Q(x|x^{(n)})$
- w.p. $P(\hat{x})/P(x^{(n)})$, $x^{(n+1)} = \hat{x}$ else $x^{(n+1)} = x^{(n)}$

Application to posterior sampling:

The denominator cancels out, leading to:

$$\frac{\beta(\theta' \mid D)}{\beta(\theta \mid D)} = \frac{P_{\theta'}(D)\beta(\theta')}{P_{\theta}(D)\beta(\theta)}$$

The only question is which proposal to use.

Metropolis-Hastings

Algorithm

- Input: Proposal distribution $Q(x|x')$ satisfying detailed balance, likelihood P.
- At time n:
- $\hat{x}|x^{(n)} \sim Q(x|x^{(n)})$
- With probability

$$\frac{P(\hat{x})Q(x^{(n)}|\hat{x})}{P(x^{(n)})Q(\hat{x}|x^{(n)})},$$

  set $x^{(n+1)} = \hat{x}$
- Otherwise $x^{(n+1)} = x^{(n)}$

Application to posterior sampling:

The $\mathbb{P}_\beta(D)$ term cancels out, leading to:

$$\frac{\beta(\theta' \mid D)Q(\theta \mid \theta')}{\beta(\theta \mid D)Q(\theta' \mid \theta)} = \frac{P_{\theta'}(D)\beta(\theta')Q(\theta \mid \theta')}{P_\theta(D)\beta(\theta)Q(\theta' \mid \theta)}$$

M-H Theory

Stationary distribution

The Markov chain defined by the M-H algorithm must have a unique stationary distribution

$$\sigma = \sigma \boldsymbol{P},$$

where $\boldsymbol{P}$ is the transition kernel of the chain with

$$P_{ij} = \mathbb{P}(x^{(n+1)} = j \mid x^{(n)} = i).$$

In addition, $\lim_{n\to\infty} \boldsymbol{P}^k = 1\sigma$.

Sufficient conditions

- If the transition kernel satisfies detailed balance:

$$P(x'|x)\sigma(x) = P(x|x')\sigma(x')$$

then $\sigma(x)$ is a stationary distribution.

- If the Markov chain is ergodic then there is a unique σ.

The Gibbs sampler

This is used when we need to sample over only some variables $z_1, \ldots, z_k$, given some fixed variables x.

General algorithm

- Input: Factors $P(z_k \mid z_1, \ldots z_{k-1}, z_{k+1}, \ldots, z_K, x)$
- For $n \in [N]$:
- For $k \in [K]$: $z_k^{(n)} \sim P(z_k \mid z_1^{(n)}, \ldots z_{k-1}^{(n)}, z_{k+1}^{(n-1)}, \ldots, z_K^{(n-1)}, x)$

Application to posterior sampling with latent variables:

Latent variable z, parameter θ.

- Until convergence:
- $\theta^{(n)} \sim P(\theta \mid z^{(n-1)}, x)$
- $z^{(n)} \sim P(z \mid \theta^{(n)}, x)$

ABC: Approximate Bayesian Computation

When to use

- When we can sample from $P_\theta(D)$.
- When we cannot calculate $P_\theta(D)$.

A metric ρ over datasets

- $\rho(D, D')$ is distance between datasets.
- We can use that to define a rejection sampler

ABC Rejection Sampling

- **Input**: $\epsilon > 0$.
- Sample $\theta' \sim \beta(\theta)$
- Sample $D' \sim P_{\theta'}$.
- If $\rho(D, D') \leq \epsilon$, accept θ'

Theorem

If $\rho(D, D') = \|f(D) - f(D')\|$ and f is a sufficient statistic and $\epsilon = 0$ then ABC Rejection Sampling is exact.

Multi-platform

- STAN
- BUGS

Python

- PyMC3
- TensorFlow Probability
- PyStan
- Pyro (Torch)